

**Report
Of
“Data Analytics Internship: Customer Segmentation and E-
Commerce Analytics Using Python”**

Submitted

To

**School of Computer Science and Engineering
IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of
Bachelor of Technology in Computer Science and Engineering



By

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Batch: 2026-27

2026-27

CANDIDATE'S DECLARATION

I, Asmi Dagar, do hereby declare that the internship report titled “Data Analytics Internship: Customer Segmentation and E-Commerce Analytics Using Python” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank Skillbit Technologies for giving me the opportunity to work as a Data Analytics Intern (Intern ID: 6210) and for providing valuable exposure to real-world data analytics and machine learning workflows during the period 13th May 2026 to 12th July 2026. I am also grateful to my faculty at IILM University, School of Computer Science and Engineering, for their continuous encouragement and support throughout this internship.

I express my gratitude to my parents for their blessings.

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INTERNSHIP COMPLETION CERTIFICATE



4. PROJECT DESCRIPTION

4.1 Introduction

This report presents the work carried out during a Data Analytics Internship undertaken at Skillbit Technologies between 13th May 2026 and 12th July 2026. The internship focused on applying Python-based data analytics and machine learning techniques to solve real-world business problems. During the course of the internship, two independent, end-to-end data analytics projects were completed: (1) Customer Segmentation using K-Means Clustering, and (2) E-Commerce Data Analysis covering customer journey, conversion rate, and marketing campaign impact. Both projects followed an industry-standard data science pipeline consisting of data loading, exploratory data analysis (EDA), data preprocessing, modelling/analysis, evaluation, visualization, and business insight generation.

The internship provided practical, hands-on exposure to core data analytics skills including data wrangling with pandas and numpy, statistical visualization with matplotlib and seaborn, unsupervised machine learning with scikit-learn, and the translation of analytical results into actionable business recommendations.

4.2 Organization Profile

Skillbit Technologies is the organization at which this internship was undertaken. The organization offers a structured Data Analytics Internship Program designed to give students practical, project-based exposure to real-world data analytics and machine learning workflows. As part of this program, interns are assigned live-style projects that mirror the working pipeline followed by data analytics teams in the industry, ranging from data generation/collection and cleaning to exploratory analysis, modelling, visualization, and business reporting. The internship was undertaken under Intern ID 6210, for the Data Analytics Intern role, for a duration of two months (13th May 2026 to 12th July 2026).

4.3 Problem Statement

Project 1 — Customer Segmentation: Businesses often treat all customers uniformly, resulting in inefficient marketing spend and generic engagement strategies. There was a need to segment mall customers into meaningful groups based on their income and spending behaviour so that targeted marketing strategies could be designed for each group.

Project 2 — E-Commerce Analysis: E-commerce businesses generate large volumes of session-level data, but without proper analysis it is difficult to identify where customers drop off in the purchase funnel, which channels and devices convert best, and which marketing campaigns generate a positive return on investment. There was a need to analyze customer journeys, conversion rates, and campaign performance to identify actionable improvement areas.

4.4 Project Objectives

- To apply unsupervised machine learning (K-Means Clustering) to segment mall customers based on Annual Income and Spending Score.
- To determine the optimal number of customer clusters using the Elbow Method and validate cluster quality using the Silhouette Score.
- To profile each customer segment and propose a tailored marketing strategy for each.
- To simulate and analyze a realistic e-commerce session dataset covering the complete customer journey from page view to purchase.
- To calculate conversion rates across devices, traffic sources, and campaigns, and to identify funnel drop-off points.
- To evaluate marketing campaign performance using ROI and Cost Per Acquisition (CPA), and to generate data-driven business recommendations.

4.5 Scope of the Project

The scope of Project 1 was limited to a dataset of 200 mall customers, using Annual Income and Spending Score as clustering features, with K-Means Clustering as the sole clustering algorithm. The scope of Project 2 covered a simulated dataset of 5,000 customer sessions over a 90-day period, incorporating traffic sources, device types, and four marketing campaigns, following an event-log structure similar to Google Analytics and Meta Ads. Both projects were implemented entirely in Python using Google Colab notebooks, and the outputs were limited to visualizations, CSV exports, and a written business report; no live deployment, dashboarding tool, or production pipeline was built as part of this internship.

4.6 Technologies and Tools Used

Technology/Tool	Purpose
Python 3.x	Core programming language

Technology/Tool	Purpose
pandas	Data manipulation, cleaning, groupby, merge, pivot_table
numpy	Numerical operations and simulated data generation
matplotlib	Data visualization and multi-panel dashboards
seaborn	Statistical plots and heatmaps
scikit-learn	K-Means clustering, StandardScaler, PCA, Silhouette Score
datetime	Timestamp generation and date/time feature engineering
Google Colab	Cloud notebook environment for development and execution

4.7 System Architecture

Both projects were built around a linear, industry-standard data analytics pipeline architecture, illustrated below:

Data Loading / Generation → Data Cleaning & Preprocessing → Exploratory Data Analysis → Modelling / Metric Computation → Evaluation → Visualization → Business Insight Generation → Export of Reports & CSVs

In Project 1, the raw Mall Customers dataset was loaded, verified for missing values, scaled using StandardScaler, and passed to a K-Means model whose optimal cluster count was chosen using the Elbow Method; results were then evaluated with the Silhouette Score and reduced to two dimensions with PCA for visualization. In Project 2, session and event-level data for 5,000 sessions was generated and merged into a master table, cleaned and validated for funnel integrity, and then processed through funnel construction, conversion-rate analysis, and campaign ROI/CPA computation modules before being visualized in a consolidated KPI dashboard.

4.8 Methodology

4.8.1 Project 1 — Customer Segmentation Using K-Means Clustering

- Step 1: Data Loading & Setup — Loaded the Mall Customers dataset (200 rows × 5 columns) and verified there were zero missing values.
- Step 2: Exploratory Data Analysis — Plotted distribution histograms for Age, Income and Spending Score, gender-wise boxplots, a correlation heatmap, and an Income vs. Spending Score scatter plot to visually identify potential clusters.
- Step 3: Data Preprocessing — Selected Annual Income and Spending Score as clustering features and normalized them using StandardScaler (mean = 0, standard deviation = 1).

- Step 4: K-Means Clustering — Applied the Elbow Method for $K = 1$ to 10, identified $K = 5$ as optimal, and trained the final K-Means model to assign cluster labels.
- Step 5: Cluster Evaluation — Computed the overall Silhouette Score (0.554) and per-cluster silhouette scores, then built a complete cluster profile.
- Step 6: Visualization — Generated a colour-coded scatter plot with centroids, a cluster-size bar chart, a normalized characteristics heatmap, and a 2D PCA cluster plot.
- Step 7: Business Insights — Named and profiled each segment, proposed a marketing strategy per segment, and exported results to CSV.

4.8.2 Project 2 — E-Commerce Data Analysis

- Task 1: Data Generation & Setup — Simulated 5,000 sessions with weighted traffic sources, nested funnel events, and four marketing campaigns; merged sessions and events into a single master table.
- Task 2: Data Cleaning & Preparation — Checked for duplicates, verified funnel integrity (zero mismatches), engineered datetime features (hour, day, week, month), and applied ordered categorical encoding to funnel stages.
- Task 3: Exploratory Data Analysis — Analyzed daily session trends, traffic-source breakdown, device distribution, hour/day-of-week patterns, and order-value distribution.
- Task 4: Customer Journey & Funnel Construction — Computed core funnel counts and drop-off percentages, mapped each session's deepest stage, and compared new vs. returning customers.
- Task 5: Conversion Rate Analysis — Calculated stage-to-stage conversion rates and conversion by device, customer type, and campaign, and compiled a full KPI scorecard.
- Task 6: Marketing Campaign Impact — Attributed revenue per campaign and computed ROI and Cost Per Acquisition (CPA) for each campaign.
- Task 7: Visualization Dashboard — Built a funnel bar chart, a device $\times$ source conversion-rate heatmap, and a consolidated 6-chart KPI dashboard.

- Task 8: Business Insights & Report — Compiled all KPIs, generated a dynamic business report, and exported CSV files for stakeholders with prioritized recommendations.

4.9 Expected Outcomes

Project 1 — Customer Segmentation Results:

Metric	Value
Dataset Size	200 customers
Optimal Clusters (K)	5
Silhouette Score	0.554 (Strong)
Features Used	Annual Income, Spending Score

Five distinct customer segments were identified: Wealthy Conservatives (avg. income \$86.5k, avg. score 17.1, 19.5% of customers), Impulsive Spenders (avg. income \$26.3k, avg. score 79.4, 11.5%), Average Customers (avg. income \$55.3k, avg. score 49.6, 40.5%), Prime Targets (avg. income \$86.1k, avg. score 82.1, 11.0%), and Careful Savers (avg. income \$26.0k, avg. score 20.9, 17.5%). Key insight: income and spending score were found to have near-zero correlation — high income does not imply high spending. Prime Targets, though only 11% of customers, represent the most valuable segment for VIP retention programs, while the 40.5% “Average” segment forms the backbone for broad seasonal campaigns.

Project 2 — E-Commerce Analysis Results:

Metric	Value
Total Sessions	5,000
Unique Customers	2,257
Total Purchases	184
Total Revenue	\$13,847.92
Overall Conversion Rate	3.68%
Average Order Value	\$75.26
Best Campaign ROI	291.3% (Retargeting_Q1)
Worst Campaign ROI	-6.2% (Influencer_Push)

Key findings included a 50% session drop-off at the homepage stage — the single largest volume loss in the funnel; mobile devices, despite carrying 58% of traffic, converted 38% worse than desktop; the Retargeting_Q1 campaign delivered the best ROI at 291%, while the Influencer_Push campaign generated a net loss; and returning customers converted 46% better than new customers. Based on these findings, recommendations were made to improve the mobile checkout experience, add homepage product recommendations, reallocate budget from the underperforming influencer campaign to retargeting, and launch a loyalty rewards program for returning customers.

4.10 Certificates of Completion and Other Communication Mail Proof

The Internship Offer Letter (dated 13/05/2026) and the Certificate of Internship Completion (Certificate ID: CSB2605503, dated 13-07-2026) issued by Skillbit Technologies are attached below as proof of the internship undertaken.



Fig 4.10.1: Internship Offer Letter



Fig 4.10.2: Internship Completion Certificate

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